MILITARY SPECIFICATION

PIN, GROOVED, HEADED-LONGITUDINAL GROOVE, METRIC, GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers metric headed-grooved pins having longitudinal grooves.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-P-416 - Plating, Cadmium (Electrodeposited)
PPP-H-1581 - Hardware (Fasteners and Related Items), Packaging of

MILITARY

DOD-P-63542/1 - Pin, Grooved, Headed-Longitudinal Grooved, Round Head, Carbon Steel, Metric

STANDARDS

FEDERAL

FED-STD-66 - Steel: Chemical Composition and Hardenability

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Armament Research and Development Command, ATTN: DRDAR-TST-S Dover, NJ 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of specifications, standards, handbooks, drawings and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references other than specification sheets cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheets. In the event of any conflict between requirements of this specification and the specification sheet, the latter shall govern.

3.2 Materials. Recycled and reclaimed materials (see 3.1) shall be used to the maximum extent practicable.

3.2.1 Carbon steel. Carbon steel pins shall be cold drawn low carbon steel wire or cold finished bar. The steel shall conform to one of the 1100 or 1200 series of FED-STD-66 and shall have a minimum ultimate shear strength of 441 MPa.

3.3 Dimensions. Dimensions and tolerances for all types of pins shall conform to the dimensional requirements of the applicable specification sheet and shall apply after protective finishes.

3.4 Performance characteristics. The pins shall meet the following performance requirements when subjected to the applicable test specified in 4.8.

3.4.1 Double shear force. The pins shall be capable of withstanding the minimum double shear force specified in table I (see 4.9.3).

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TABLE I. Double shear force.

Nominal pin diameter (mm)	Double shear force (kilonewtons) min
	Carbon steel
1.4	1.3
1.7	2.0
2.0	2.8
2.3	3.7
2.6	4.8
3.0	6.3
4.0	11.3
5.0	17.6
6.0	25.4

3.5 Finish. The protective finish of the pins shall be as follows (see 4.9.2).

3.5.1 Cadmium plate. Carbon steel pins shall be cadmium plated in accordance with QQ-P-416, type I, class 3 (5.1 µm thick).

3.6 Workmanship. The workmanship shall be uniform in quality and free of irregularities or detrimental defects. The surface shall not contain any foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Materials inspection (see 4.4)
b. Quality conformance inspection (see 4.5)

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in 4.9.1, 4.9.2, and 4.9.3.

4.4 Materials inspection. Materials inspection shall consist of certification supported by verifying data that the materials used in fabricating the pins are in accordance with the requirements of 3.2 prior to such fabrication.

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4.5 Quality conformance inspection. Quality conformance inspection shall be as specified in table II. Each pin which does not pass any of the tests specified shall be defective.

TABLE II. Quality conformance inspection.

Inspection	Requirement paragraph	Test method paragraph
Group A		
Dimensions	3.3	4.9.1
Finish	3.5.1	4.9.2
Group C		
Shear strength	3.4.1	4.9.3

4.6 Quality conformance inspection sampling.

4.6.1 Inspection lot. An inspection lot shall consist of all pins covered by a single specification sheet, produced under essentially the same conditions, and offered for inspection at any one time.

4.6.2 Sampling for group A inspection. Sampling for group A inspection shall be as specified in 4.6.2.1 and 4.6.2.2.

4.6.2.1 Sampling for dimensions. Sampling shall be in accordance with MIL-STD-105 inspection level II. The Acceptable Quality Level (AQL) shall be as indicated in table III. The AQL shall apply to each individual defect, not to a group of defects.

4.6.2.2 Sampling for finish. Sampling for protective finish test of pins shall be in accordance with the applicable specification in 3.5.

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TABLE III. Classification of defects.

Category	Defect	Inspection method
Critical	None defined	
Major	AQL = 2.5 percent defective	
101	Expanded diameter of pin	Gage
102	Nominal diameter of pin	SIE 1/
103	Grooves missing or wrong type grooves	Visual
Minor	AQL = 4.0 percent defective	
201	Length of pin	SIE 1/
202	Head diameter of pin	SIE 1/
203	Protective finish missing (see 4.9.2)	Visual
204	Workmanship	Visual

1/ Standard inspection equipment

4.6.3 Sampling for group C inspection. Sampling shall be in accordance with level S-2 of MIL-STD-105. The AQL shall be 1.5 percent defective. Group C inspection shall be performed on sample units which have passed group A inspection.

4.7 Noncompliance. If sample fails to pass group C inspection, the manufacturer shall notify the qualifying activity and the cognizant inspection activity of such failure and take corrective action on the materials or processes, or both, as warranted, and on all units of product which can be corrected and which are manufactured under essentially the same materials and processes, and which are considered subject to the same failure. Acceptance and shipment of the product shall be discontinued until corrective action, acceptable to the qualifying activity has been taken. After the corrective action has been taken, group C inspection shall be repeated on additional sample units (all tests and examinations, or the test which the original sample failed, at the option of the qualifying activity). Group A inspections may be reinstituted; however, final acceptance and shipment shall be withheld until the group C inspection has shown that the corrective action was successful. In the event of failure after reinspection, information concerning the failure shall be furnished to the cognizant inspection activity and the qualifying activity.

4.8 Inspection of packaging. The sampling and inspection of the preservation-packaging, packing, and container marking shall be in accordance with the requirements of PPP-H-1581.

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4.9 Methods of inspection.

4.9.1 Dimensional examination. Each pin taken as specified in 4.6.2.1 shall be examined to verify conformance with this specification. Examination shall be conducted in accordance with table III.

4.9.2 Finish test. The pins shall conform to the finish test of QQ-P-416, as specified in 3.5.

4.9.3 Shear strength test. Each pin taken as specified in 4.6.3 shall be tested for shear strength in accordance with 4.9.3.1.

4.9.3.1 Shear test fixture. The shear test shall be made by means of a suitable fixture which meets the following requirements. The shear plane shall be at least one pin diameter away from the end of the pin. Clearance between the loading and supporting members shall be 0.127 mm maximum. Hardness of the loading and supporting members shall be 55HRC minimum. The hole for the pins shall have sharp edges and shall be as specified in the applicable specification sheet.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with PPP-H-1581 (see 6.2).

6. NOTES

6.1 Intended use. Pins covered by this specification are intended for use in such application where threaded fasteners might be used, but where use of headed groove pins afford rapid application and greater economy.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification and applicable specification sheet.
b. Applicable military specification sheet part number(s).
c. Finish (see 3.5.1).
d. Applicable levels of packaging (see 5.1).

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